University of Florida, Department of Mathematics

Algebra First-Year Exam, May 2013, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let R be an integral domain, let $P \subset R$ be a prime ideal, and let $F = \mathrm{Frac}(R)$ be the field of fractions of R. Define

$$R_P = \left\{ \frac{a}{b} : a, b \in R,\ b \notin P \right\}.$$

(a) Prove that R_P is a subring of F.

(b) Prove that R_P has a unique maximal ideal.

2. Let $\omega = -\frac{1}{2} + \frac{1}{2}\sqrt{-3} \in \mathbb{C}$.

(a) Prove that ω is a root of the polynomial $f(X) = X^2 + X + 1$.

(b) Prove that the set $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{C}$.

(c) Prove that $\mathbb{Z}[\omega]$ is a Euclidean domain with respect to the norm $N(\alpha) = \alpha\overline{\alpha}$, where $\overline{\alpha}$ denotes the complex conjugate of $\alpha \in \mathbb{Z}[\omega]$.

3. Let R be an integral domain and let $P \subset R$ be a prime ideal. Let

$$f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0$$

be a monic polynomial of degree $n \geq 1$ with coefficients in R.

(a) Prove Eisenstein's criterion: If $a_i \in P$ for $0 \leq i \leq n-1$ and $a_0 \notin P^2$, then $f(X)$ is irreducible in $R[X]$.

(b) Give an example of $f(X)$ as above such that $a_i \in P$ for $0 \leq i \leq n-1$ but $f(X)$ is not irreducible.

4. This problem has three parts.

(a) Let R be an integral domain and let M be an R-module. Define the rank of M.

(b) Prove that $\mathbb{Q}$ has rank 1 as a $\mathbb{Z}$-module.

(c) Prove that $\mathbb{Q}$ is not a free $\mathbb{Z}$-module.

5. Find a representative for each similarity class of noninvertible 3×3 matrices with entries in the field $\mathbb{F}_2$ with 2 elements.

6. Let E/F be a field extension and let R be a subring of E which contains F.

(a) Prove that if E/F is an algebraic extension then R is a field.

(b) Give an example where E/F is not an algebraic extension and R is not a field.